

## 1 Chapter 2.1

**Problem 4.** Consider the possible functions  $f : [7] \rightarrow [9]$

- (a) How many have  $f(3) = 8$ ? How many have  $f(3) \neq 8$ ?
- (b) How many have  $f(1) \neq 5$  and are one-to-one?
- (c) How many have  $f(i)$  even, for all  $i$ ?
- (d) How many have  $\text{rng}(f) = \{5, 6\}$ ?
- (e) How many in which  $f^{-1}$  is not a function?

**Problem 7.** Find the number of onto functions  $[k] \rightarrow [4]$ .

**Problem 11.** Give a combinatorial proof. For  $n \geq 1$  and  $k \geq 1$ , then  $2^{kn} > \max\{n^k, k^n\}$ . *Hint: Compare relations to functions.*

## 2 Chapter 2.2

**Problem 4.** Only include  $a$ ,  $b$ ,  $d$

**Problem 5.** Rawr

**Problem 7.** Only include  $a$ ,  $b$ ,  $c$

## 3 Chapter 2.3

**Problem 2.** Rawr

**Problem 7.** Rawr